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Mathias Marty

Education

2022–2024 **Master of Science - Cyber Security**, *EPFL/ETHZ*, Lausanne/Zürich, GPA: *5.34/6*

2021–2022 **HES–EPFL transition program**, *EPFL*, Lausanne, GPA: *5.12/6*

2018–2021 **Bachelor of Science - Computer Science**, *HES/SO Haute Ecole Arc*, Neuchâtel, GPA: *5.24/6*

2015–2018 **Federal Diploma of Vocational Education and Training (Computer Science) and Professional Baccalaureate**, *CPNE*, Neuchâtel

Publications

Under review / In preparation

[1] Jonathan Bootle, Alessandro Chiesa, and Mathias Marty. Round-by-round knowledge soundness for linear-time iops. 2026. Submitted to Crypto 2026.

[2] Iftach Haitner, Mathias Marty, and Eylon Yogev. Lower bound for the size of hash based signatures. 2026. Manuscript in preparation.

Published / Accepted Papers

[3] Antoine Lestrade, Mathias Marty, Artan Sadiku, Christophe Muller, Joep Neijt, Yann Voumard, and Stéphane Gobron. Real-time renderings of multidimensional massive datacubes on jupyter notebook. In Liang-Yee Cheng, editor, *Proceedings of the 20th International Conference on Geometry and Graphics*, ICGG '22, pages 685–696. Springer International Publishing, 2023. [link].

[4] Mathias Marty, Antoine Lestrade, Artan Sadiku, Christophe Muller, Joep Neijt, Yann Voumard, and Stéphane Gobron. Implicit curves: From discrete extraction to applied formalism. In *Proceedings of the 20th International Conference on Geometry and Graphics*, ICGG '22, pages 717–728, 2022. [link].

[5] Christophe Muller, Antoine Lestrade, Mathias Marty, Artan Sadiku, Joep Neijt, Yann Voumard, and Stéphane Gobron. Earth observation datacubes multi-visualization toolbox. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-4/W1-2022, 2022. [link].

Research projects

2024–present **Independent Research**, *with Prof. Alessandro Chiesa and Dr. Jonathan Bootle*
Round-by-Round Knowledge Soundness for Linear-Time IOPs

2025–present **Independent Research**, *with Prof. Eylon Yogev and Prof. Iftach Haitner*
Implementing a PoC in Rust and lower bounds for hash-based signatures

2024 **Master Project**, *supervised by Prof. Alessandro Chiesa*, EPFL, grade: *5.75/6*
On State-Restoration Knowledge Soundness from Special Soundness (pdf, presentation)

2023 **Semester Project**, *supervised by Prof. Serge Vaudenay*, EPFL, grade: *5.75/6*
Code-based Cryptography in the Lee metric (pdf, presentation)

2022 **Formal Verification Course Project**, *EPFL*, not graded
Prove some RSA security properties with the proof assistant EasyCrypt

2021 **Bachelor Project**, *supervised by Prof. Stéphane Gobron*, HES/SO Haute Ecole Arc, grade: *6/6*
DataCube: Implicit Curves (pdf (french only))

Skills

Practical In addition to my theoretical skills, I acquired many practical skills.

- **Programming:** Rust, Assembly, C, VHDL, C++, Scala & Stainless, Java, C#, JavaScript, WebGL/GLSL, SQL, Python/Sage, Kotlin
- **Software and tools:** Git, Linux, EasyCrypt, GDB, CUDA, LaTeX, QEMU, RTOS, embedded devices, Docker, American Fuzzy Lop
- **Soft skills:** Writing technical reports and papers, making presentations, teaching, ability to adapt to different learning levels and needs, communication

Related Courses Cryptography and Security (*5*), Advanced Cryptography (*5.25*), Number Theory in Cryptography (*5.5*), Foundations of Probabilistic Proofs (*5*), Computational Complexity (*5.25*), Computer Architecture (*5*), Operating Systems (*5.5*), Computer Networks (*5.75*), Algorithms (*5*), Advanced Algorithms (*5.25*), Formal Verification (*5*), Gödel and Recursivity (*5.75*), Advanced Encryption Schemes, Zero-Knowledge Proofs...

Experience

Feb–Jul 2026 (upcoming) **Software Engineering Internship**, *Corintis*, Lausanne
Software engineer at Corintis, developing software for chip-level optimized cooling systems

2025 **Programmer**, *Prof. Eylon Yogev*, 50% work for Prof. Eylon Yogev
Implementing cryptographic primitives and proofs of concepts

2023 **Summer internship**, *Swiss Center for Electronics and Microtechnology (CSEM)*, Neuchâtel, supervised by Dr. Damian Vizár
Embedded fuzzing testbench of a RTOS with HALucinator *evaulation: Excellent*

2018–present **Private tutoring**
Tutored students across a wide academic range, from elementary to University level (I now have a Superprof account)

Languages

French (native), English (fluent; TOEFL *102/120* in 2024, FCE in 2018)